Asymmetric weak Polynomial Freiman-Ruzsa over the integers

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Apache-2.0

Editorial note. Source-organized author material only. All mathematical source spans are retained unchanged. Blueprint lean/proves/uses metadata is hidden as formatting only; numbering is local to this material wrapper.

Editorial note. Ordinary cited prerequisites and concise author arguments are accepted as such. No independent mathematical certification or public ID allocation is claimed.

Allowed standard prerequisites (editorial).

- Standard finite Shannon entropy identities, conditional entropy and finite abelian group algebra are cited in their original Blueprint formulations.
- Named prior PFR/entropy or BSG results are allowed prerequisites; their original statements are retained where used, without demanding recursively expanded proofs.

1 The Fibring lemma

Proposition 1 (General fibring identity)**.** *Let $\pi : H \to H'$ be a homomorphism additive groups, and let Z_1, Z_2 be H-valued random variables. Then we have*

$$d[Z_1; Z_2] \geq d[\pi(Z_1); \pi(Z_2)] + d[Z_1|\pi(Z_1); Z_2|\pi(Z_2)].$$

Moreover, if Z_1, Z_2 are taken to be independent, then the difference between the two sides is

$$I(Z_1 - Z_2 : (\pi(Z_1), \pi(Z_2))|\pi(Z_1 - Z_2)).$$

Proof. Let Z_1, Z_2 be independent throughout (this is possible by Theorem 28 and Theorem 25). By Theorem 36, We have

$$\begin{aligned}
&d[Z_1|\pi(Z_1); Z_2|\pi(Z_2)] \\
&= \mathbb{H}[Z_1 - Z_2|\pi(Z_1), \pi(Z_2)] - \tfrac{1}{2}\mathbb{H}[Z_1|\pi(Z_1)] - \tfrac{1}{2}\mathbb{H}[Z_2|\pi(Z_2)] \\
&\leq \mathbb{H}[Z_1 - Z_2|\pi(Z_1 + Z_2)] - \tfrac{1}{2}\mathbb{H}[Z_1|\pi(Z_1)] - \tfrac{1}{2}H[Z_2|\pi(Z_2)] \\
&= d[Z_1; Z_2] - d[\pi(Z_1); \pi(Z_2)].
\end{aligned}$$

In the middle step, we used Theorem 14, and in the last step we used the fact that

$$\mathbb{H}[Z_1 - Z_2|\pi(Z_1 - Z_2)] = \mathbb{H}[Z_1 - Z_2] - \mathbb{H}[\pi(Z_1 - Z_2)]$$

(thanks to Theorem 9 and Theorem 4) and that

$$\mathbb{H}[Z_i|\pi(Z_i)] = \mathbb{H}[Z_i] - \mathbb{H}[\pi(Z_i)]$$

(since Z_i determines $\pi(Z_i)$). This gives the claimed inequality. The difference between the two sides is precisely

$$\mathbb{H}[Z_1 - Z_2|\pi(Z_1 - Z_2)] - \mathbb{H}[Z_1 - Z_2|\pi(Z_1), \pi(Z_2)].$$

To rewrite this in terms of (conditional) mutual information, we use the identity

$$\mathbb{H}[A|B] - \mathbb{H}[A|B, C] = \mathbb{I}[A : C|B],$$

(which follows Theorem 18) taking $A := Z_1 - Z_2$, $B := \pi(Z_1 - Z_2)$ and $C := (\pi(Z_1), \pi(Z_2))$, and noting that in this case $\mathbb{H}[A|B, C] = \mathbb{H}[A|C]$ since C uniquely determines B (this may require another helper lemma about entropy). This completes the proof. □

Corollary 2. *If $\pi : G \to H$ is a homomorphism of additive groups and X, Y are G-valued random variables then*

$$d[X; Y] \geq d[\pi(X); \pi(Y)].$$

Proof. By Theorem 1 and the nonnegativity of conditional Ruzsa distance (from Theorem 31) we have

$$d[X; Y] \geq d[\pi(X); \pi(Y)] + d[X \mid \pi(X); Y \mid \pi(Y)].$$

The inequality follows from $d[X \mid \pi(X); Y \mid \pi(Y)] \geq 0$ (Theorem 31). □

2 Shannon entropy inequalities

Random variables in this paper are measurable maps $X : \Omega \to S$ from a probability space Ω to a measurable space S, and called S-valued random variables. In many cases we will assume that singletons in S are measurable. Often we will restrict further to the case when S is finite with the discrete σ-algebra, which of course implies that S has measurable singletons.

Definition 3 (Entropy)**.** *If X is an S-valued random variable, the entropy $\mathbb{H}[X]$ of X is defined*

$$\mathbb{H}[X] := \sum_{s \in S} \mathbb{P}[X = x] \log \frac{1}{\mathbb{P}[X = x]}$$

with the convention that $0 \log \frac{1}{0} = 0$.

Lemma 4 (Entropy and relabeling)**.** *(i) If $X : \Omega \to S$ and $Y : \Omega \to T$ are random variables, and $Y = f(X)$ for some injection $f : S \to T$, then $\mathbb{H}[X] = \mathbb{H}[Y]$.*

(ii) If $X : \Omega \to S$ and $Y : \Omega \to T$ are random variables, and $Y = f(X)$ and $X = g(Y)$ for some functions $f : S \to T$, $g : T \to S$, then $\mathbb{H}[X] = \mathbb{H}[Y]$.

Proof. Expand out both entropies and rearrange. □

Lemma 5 (Jensen bound)**.** *If X is an S-valued random variable, then $\mathbb{H}[X] \leq \log |S|$.*

Proof. This is a direct consequence of Theorem 19 and Jensen's inequality. □

We use X, Y to denote the pair $\omega \mapsto (X(\omega), Y(\omega))$.

Lemma 6 (Commutativity and associativity of joint entropy)**.** *If $X : \Omega \to S$, $Y : \Omega \to T$, and $Z : \Omega \to U$ are random variables, then $\mathbb{H}[X, Y] = \mathbb{H}[Y, X]$ and $\mathbb{H}[X, (Y, Z)] = \mathbb{H}[(X, Y), Z]$.*

Proof. Set up an injection from (X, Y) to (Y, X) and use Theorem 4 for the first claim. Similarly for the second claim. □

Note: it may happen that E has zero measure. In which case, the ambient probability measure should be replaced with a zero measure. (In our formalization we achieve this by working with arbitrary measures, and normalizing them to be probability measures if possible, else using the zero measure. Conditioning is also formalized using existing Mathlib definitions.)

Definition 7 (Conditional entropy)**.** *If $X : \Omega \to S$ and $Y : \Omega \to T$ are random variables, the conditional entropy $\mathbb{H}[X|Y]$ is defined as*

$$\mathbb{H}[X|Y] := \sum_{y \in Y} \mathbb{P}[Y = y] \mathbb{H}[(X|Y = y)].$$

Lemma 8 (Conditional entropy and relabeling)**.** *If $X : \Omega \to S$, $Y : \Omega \to T$, and $Z : \Omega \to U$ are random variables, and $Y = f(X, Z)$ almost surely for some map $f : S \times U \to T$ that is injective for each fixed U, then $\mathbb{H}[X|Z] = \mathbb{H}[Y|Z]$.*

Similarly, if $g : T \to U$ is injective, then $\mathbb{H}[X|g(Y)] = \mathbb{H}[X|Y]$.

Proof. For the first part, use Theorem 7 and then Theorem 4. The second part is a direct computation. □

Lemma 9 (Chain rule)**.** *If $X : \Omega \to S$ and $Y : \Omega \to T$ are random variables, then*

$$\mathbb{H}[X, Y] = \mathbb{H}[Y] + \mathbb{H}[X|Y].$$

Proof. Direct computation. □

Lemma 10 (Alternative formulae for mutual information)**.** *With notation as above, we have*

$$\mathbb{I}[X : Y] = \mathbb{I}[Y : X]$$

$$\mathbb{I}[X : Y] = \mathbb{H}[X] - \mathbb{H}[X|Y]$$

$$\mathbb{I}[X : Y] = \mathbb{H}[Y] - \mathbb{H}[Y|X]$$

Proof. Immediate from Lemmas 6, 9. □

Lemma 11 (Nonnegativity of mutual information)**.** *We have $\mathbb{I}[X : Y] \geq 0$.*

Proof. An application of jensen's inequality and Theorems 10 and 19. □

Corollary 12 (Subadditivity)**.** *With notation as above, we have* $\mathbb{H}[X,Y] \leq \mathbb{H}[X] + \mathbb{H}[Y]$.

Proof. Use Theorem 11. □

Corollary 13 (Conditioning reduces entropy)**.** *With notation as above, we have* $\mathbb{H}[X|Y] \leq \mathbb{H}[X]$.

Proof. Combine Theorem 11 with Theorem 10. □

Corollary 14 (Submodularity)**.** *With three random variables* X, Y, Z*, one has* $\mathbb{H}[X|Y,Z] \leq \mathbb{H}[X|Z]$.

Proof. Apply the "averaging over conditioning" argument to Theorem 13. □

Corollary 15 (Alternate form of submodularity)**.** *With three random variables* X, Y, Z*, one has*

$$\mathbb{H}[X,Y,Z] + \mathbb{H}[Z] \leq \mathbb{H}[X,Z] + \mathbb{H}[Y,Z].$$

Proof. Apply Theorem 14 and Theorem 9. □

Lemma 16 (Vanishing of mutual information)**.** *If* X, Y *are random variables, then* $\mathbb{I}[X:Y] = 0$ *if and only if* X, Y *are independent.*

Proof. An application of the equality case of Jensen's inequality and Theorem 19. □

Corollary 17 (Additivity of entropy)**.** *If* X, Y *are random variables, then* $\mathbb{H}[X,Y] = \mathbb{H}[X] + \mathbb{H}[Y]$ *if and only if* X, Y *are independent.*

Proof. Direct from Theorem 16. □

Lemma 18 (Alternate formula for conditional mutual information)**.** *We have*

$$\mathbb{I}[X:Y|Z] := \mathbb{H}[X|Z] + \mathbb{H}[Y|Z] - \mathbb{H}[X,Y|Z]$$

and

$$\mathbb{I}[X:Y|Z] := \mathbb{H}[X|Z] - \mathbb{H}[X|Y,Z].$$

Proof. Routine computation. □

3 Applications of Jensen's inequality

In this chapter, h denotes the function $h(x) := x \log \frac{1}{x}$ for $x \in [0,1]$.

Lemma 19 (Concavity)**.** h *is strictly concave on* $[0,\infty)$.

Proof. Check that h' is strictly monotone decreasing. □

4 Entropic Ruzsa calculus

In this section G will be a finite additive group. (May eventually want to generalize to infinite G.)

Lemma 20 (Negation preserves entropy)**.** *If X is G-valued, then $\mathbb{H}[-X] = \mathbb{H}[X]$.*

Proof. Immediate from Theorem 4. □

Lemma 21 (Shearing preserves entropy)**.** *If X, Y are G-valued, then $\mathbb{H}[X \pm Y|Y] = \mathbb{H}[X|Y]$ and $\mathbb{H}[X \pm Y, Y] = \mathbb{H}[X, Y]$.*

Proof. Immediate from Theorem 8 and Theorem 9. □

Lemma 22 (Lower bound of sumset)**.** *If X, Y are G-valued random variables on Ω, we have*

$$\max(\mathbb{H}[X], \mathbb{H}[Y]) - \mathbb{I}[X : Y] \leq \mathbb{H}[X \pm Y].$$

Proof. By Theorem 13, 21, 10, 20 we have

$$\mathbb{H}[X \pm Y] \geq \mathbb{H}[X \pm Y|Y] = \mathbb{H}[X|Y] = \mathbb{H}[X] - \mathbb{I}[X : Y]$$

and similarly with the roles of X, Y reversed, giving the claim. □

Corollary 23 (Independent lower bound on sumset)**.** *If X, Y are independent G-valued random variables, then*

$$\max(\mathbb{H}[X], \mathbb{H}[Y]) \leq \mathbb{H}[X \pm Y].$$

Proof. Combine Theorem 22 with Theorem 16. □

One random variable is said to be a copy of another if they have the same distribution.

Lemma 24 (Copy preserves entropy)**.** *If X' is a copy of X then $\mathbb{H}[X'] = \mathbb{H}[X]$.*

Proof. Immediate from Theorem 3. □

Lemma 25 (Existence of independent copies)**.** *Let $X_i : \Omega_i \to S_i$ be random variables for $i = 1, \ldots, k$. Then if one gives $\prod_{i=1}^k S_i$ the product measure of the laws of X_i, the coordinate functions $(x_j)_{j=1}^k \mapsto x_i$ are jointly independent random variables which are copies of the $X_1, \ldots, X_k$.*

Proof. Explicit computation. □

Definition 26 (Ruzsa distance)**.** *Let X, Y be G-valued random variables (not necessarily on the same sample space). The* Ruzsa distance *$d[X; Y]$ between X and Y is defined to be*

$$d[X; Y] := \mathbb{H}[X' - Y'] - \mathbb{H}[X']/2 - \mathbb{H}[Y']/2$$

where X', Y' are (the canonical) independent copies of X, Y from Theorem 25.

Lemma 27 (Distance from zero)**.** *If X is a G-valued random variable and 0 is the random variable taking the value 0 everywhere then*

$$d[X;0] = \mathbb{H}(X)/2.$$

Proof. This is an immediate consequence of the definitions and $X - 0 \equiv X$ and $\mathbb{H}(0) = 0$. □

Lemma 28 (Copy preserves Ruzsa distance)**.** *If X', Y' are copies of X, Y respectively then $d[X';Y'] = d[X;Y]$.*

Proof. Immediate from Definitions 26 and Theorem 24. □

Lemma 29 (Ruzsa distance in independent case)**.** *If X, Y are independent G-random variables then*

$$d[X;Y] := \mathbb{H}[X - Y] - \mathbb{H}[X]/2 - \mathbb{H}[Y]/2.$$

Proof. Immediate from Theorem 26 and Lemmas 4, 24. □

Lemma 30 (Distance controls entropy difference)**.** *If X, Y are G-valued random variables, then*

$$|\mathbb{H}[X] - H[Y]| \leq 2d[X;Y].$$

Proof. Immediate from Theorem 23 and Theorem 26, and also Theorem 20. □

Lemma 31 (Distance nonnegative)**.** *If X, Y are G-valued random variables, then*

$$d[X;Y] \geq 0.$$

Proof. Immediate from Theorem 30. □

Lemma 32 (Projection entropy and distance)**.** *If G is an additive group and X is a G-valued random variable and $H \leq G$ is a finite subgroup then, with $\pi : G \to G/H$ the natural homomorphism we have (where U_H is uniform on H)*

$$\mathbb{H}(\pi(X)) \leq 2d[X;U_H].$$

Proof. WLOG, we make X, U_H independent (Theorem 25). Now by Lemmas 14, 21, 5

$$\begin{aligned} \mathbb{H}(X - U_H|\pi(X)) &\geq \mathbb{H}(X - U_H|X) &&= \mathbb{H}(U_H) \\ \mathbb{H}(X - U_H|\pi(X)) &\leq \log|H| &&= \mathbb{H}(U_H) \end{aligned}$$

By Theorem 9

$$\mathbb{H}(X - U_H) = \mathbb{H}(\pi(X)) + \mathbb{H}(X - U_H|\pi(X)) = \mathbb{H}(\pi(X)) + \mathbb{H}(U_H)$$

and therefore

$$d[X;U_H] = \mathbb{H}(\pi(X)) + \frac{\mathbb{H}(U_H) - \mathbb{H}(X)}{2}.$$

Furthermore by Theorem 30

$$d[X;U_H] \geq \frac{|\mathbb{H}(X) - \mathbb{H}(U_H)|}{2}.$$

Adding these inequalities gives the result. □

Lemma 33 (Improved Ruzsa triangle inequality)**.** *If X, Y, Z are G-valued random variables on Ω with (X, Y) independent of Z, then*

$$\mathbb{H}[X - Y] \leq \mathbb{H}[X - Z] + \mathbb{H}[Z - Y] - \mathbb{H}[Z] \tag{1}$$

This is an improvement over the usual Ruzsa triangle inequality because X, Y are not assumed to be independent. However we will not utilize this improvement here.

Proof. Apply Theorem 15 to obtain

$$\mathbb{H}[X - Z, X - Y] + \mathbb{H}[Y, X - Y] \geq \mathbb{H}[X - Z, Y, X - Y] + \mathbb{H}[X - Y].$$

Using

$$\mathbb{H}[X - Z, X - Y] \leq \mathbb{H}[X - Z] + \mathbb{H}[Y - Z]$$

(from Theorem 4, Theorem 12),

$$\mathbb{H}[Y, X - Y] = \mathbb{H}[X, Y]$$

(from Theorem 4), and

$$\mathbb{H}[X - Z, Y, X - Y] = \mathbb{H}[X, Y, Z] = \mathbb{H}[X, Y] + \mathbb{H}[Z]$$

(from Theorem 4 and Theorem 17) and rearranging, we indeed obtain (1). □

Lemma 34 (Ruzsa triangle inequality)**.** *If X, Y, Z are G-valued random variables, then*

$$d[X; Y] \leq d[X; Z] + d[Z; Y].$$

Proof. By Theorem 28 and Lemmas 25, 29, it suffices to prove this inequality assuming that X, Y, Z are defined on the same space and are independent. But then the claim follows from Theorem 33 and Theorem 26. □

Definition 35 (Conditioned Ruzsa distance)**.** *If (X, Z) and (Y, W) are random variables (where X and Y are G-valued) we define*

$$d[X|Z; Y|W] := \sum_{z,w} \mathbb{P}[Z = z]\mathbb{P}[W = w] d[(X|Z = z); (Y|(W = w))].$$

similarly

$$d[X; Y|W] := \sum_{w} \mathbb{P}[W = w] d[X; (Y|(W = w))].$$

Lemma 36 (Alternate form of distance)**.** *The expression $d[X|Z; Y|W]$ is unchanged if (X, Z) or (Y, W) is replaced by a copy. Furthermore, if (X, Z) and (Y, W) are independent, then*

$$d[X|Z; Y|W] = \mathbb{H}[X - Y|Z, W] - \mathbb{H}[X|Z]/2 - \mathbb{H}[Y|W]/2$$

and similarly

$$d[X; Y|W] = \mathbb{H}[X - Y|W] - \mathbb{H}[X]/2 - \mathbb{H}[Y|W]/2.$$

Proof. Straightforward thanks to Theorem 24, Theorem 28, Theorem 29, Theorem 35, Theorem 7. □

Theorem 37 (Improved entropy version of PFR)**.** *Let $G = \mathbb{F}_2^n$, and suppose that X_1^0, X_2^0 are G-valued random variables. Then there is some subgroup $H \leq G$ such that*

$$d[X_1^0; U_H] + d[X_2^0; U_H] \leq 10 d[X_1^0; X_2^0],$$

where U_H is uniformly distributed on H. Furthermore, both $d[X_1^0; U_H]$ and $d[X_2^0; U_H]$ are at most $\frac{11}{2} d[X_1^0; X_2^0]$.

5 Weak PFR over the integers

Lemma 38. *If G is torsion-free and X, Y are G-valued random variables then $d[X; 2Y] \leq 5d[X; Y]$.*

Proof. Let Y_1, Y_2 be independent copies of Y (also independent of X). Since G is torsion-free we know $X, Y_1 - Y_2, X - 2Y_1$ uniquely determine X, Y_1, Y_2 and so

$$\mathbb{H}(X, Y_1, Y_2, X - 2Y_1) = \mathbb{H}(X, Y_1, Y_2) = \mathbb{H}(X) + 2\mathbb{H}(Y).$$

Similarly

$$\mathbb{H}(X, X - 2Y_1) = \mathbb{H}(X) + \mathbb{H}(2Y_1) = \mathbb{H}(X) + \mathbb{H}(Y).$$

Furthermore

$$\mathbb{H}(Y_1 - Y_2, X - 2Y_1) = \mathbb{H}(Y_1 - Y_2, X - Y_1 - Y_2) \leq \mathbb{H}(Y_1 - Y_2) + \mathbb{H}(X - Y_1 - Y_2).$$

By submodularity (Theorem 15)

$$\mathbb{H}(X, Y_1, Y_2, X - 2Y_1) + \mathbb{H}(X - 2Y_1) \leq \mathbb{H}(X, X - 2Y_1) + \mathbb{H}(Y_1 - Y_2, X - 2Y_1).$$

Combining these inequalities

$$\mathbb{H}(X - 2Y_1) \leq \mathbb{H}(Y_1 - Y_2) + \mathbb{H}(X - Y_1 - Y_2) - \mathbb{H}(Y).$$

Similarly we have

$$\mathbb{H}(Y_1, Y_2, X - Y_1 - Y_2) = \mathbb{H}(X) + 2\mathbb{H}(Y),$$

$$\mathbb{H}(Y_1, X - Y_1 - Y_2) = \mathbb{H}(Y) + \mathbb{H}(X - Y_2),$$

and

$$\mathbb{H}(Y_2, X - Y_1 - Y_2) = \mathbb{H}(Y) + \mathbb{H}(X - Y_1)$$

and by submodularity (Theorem 15) again

$$\mathbb{H}(Y_1, Y_2, X - Y_1 - Y_2) + \mathbb{H}(X - Y_1 - Y_2) \leq \mathbb{H}(Y_1, X - Y_1 - Y_2) + \mathbb{H}(Y_2, X - Y_1 - Y_2).$$

Combining these inequalities (and recalling the definition of Ruzsa distance) gives

$$\mathbb{H}(X - Y_1 - Y_2) \leq \mathbb{H}(X - Y_1) + \mathbb{H}(X - Y_2) - \mathbb{H}(X) = 2d[X; Y] + \mathbb{H}(Y).$$

It follows that

$$\mathbb{H}(X - 2Y_1) \leq \mathbb{H}(Y_1 - Y_2) + 2d[X; Y]$$

and so (using $\mathbb{H}(2Y) = \mathbb{H}(Y)$)

$$\begin{aligned} d[X;2Y] &= \mathbb{H}(X - 2Y_1) - \mathbb{H}(X)/2 - \mathbb{H}(2Y)/2 \\ &\leq \mathbb{H}(Y_1 - Y_2) + 2d[X;Y] - \mathbb{H}(X)/2 - \mathbb{H}(Y)/2 \\ &= d[Y_1;Y_2] + \frac{\mathbb{H}(Y) - \mathbb{H}(X)}{2} + 2d[X;Y]. \end{aligned}$$

Finally note that by the triangle inequality (Theorem 34) we have

$$d[Y_1;Y_2] \leq d[Y_1;X] + d[X;Y_2] = 2d[X;Y].$$

The result follows from $(\mathbb{H}(Y) - \mathbb{H}(X))/2 \leq d[X;Y]$ (Theorem 30). □

Lemma 39. *If G is a torsion-free group and X, Y are G-valued random variables and $\phi : G \to \mathbb{F}_2^d$ is a homomorphism then*

$$\mathbb{H}(\phi(X)) \leq 10d[X;Y].$$

Proof. By Theorem 2 and Theorem 38 we have

$$d[\phi(X);\phi(2Y)] \leq d[X;2Y] \leq 5d[X;Y]$$

and $\phi(2Y) = 2\phi(Y) \equiv 0$ so the left-hand side is equal to $d[\phi(X);0] = \mathbb{H}(\phi(X))/2$ (using Theorem 27). □

Lemma 40. *Let $G = \mathbb{F}_2^n$ and $\alpha \in (0,1)$ and let X, Y be G-valued random variables such that*

$$\mathbb{H}(X) + \mathbb{H}(Y) > \frac{20}{\alpha} d[X;Y].$$

There is a non-trivial subgroup $H \leq G$ such that

$$\log|H| < \frac{1+\alpha}{2}(\mathbb{H}(X) + \mathbb{H}(Y))$$

and

$$\mathbb{H}(\psi(X)) + \mathbb{H}(\psi(Y)) < \alpha(\mathbb{H}(X) + \mathbb{H}(Y))$$

where $\psi : G \to G/H$ is the natural projection homomorphism.

Proof. By Theorem 37 there exists a subgroup H such that $d[X;U_H] + d[Y;U_H] \leq 10d[X;Y]$. Using Theorem 32 we deduce that $\mathbb{H}(\psi(X)) + \mathbb{H}(\psi(X)) \leq 20d[X;Y]$. The second claim follows adding these inequalities and using the assumption on $\mathbb{H}(X) + \mathbb{H}(Y)$.

Furthermore we have by Theorem 30

$$\log|H| - \mathbb{H}(X) \leq 2d[X;U_H]$$

and similarly for Y and thus

$$\begin{aligned} \log|H| &\leq \frac{\mathbb{H}(X) + \mathbb{H}(Y)}{2} + d[X;U_H] + d[Y;U_H] \leq \frac{\mathbb{H}(X) + \mathbb{H}(Y)}{2} + 10d[X;Y] \\ &< \frac{1+\alpha}{2}(\mathbb{H}(X) + \mathbb{H}(Y)). \end{aligned}$$

Finally note that if H were trivial then $\psi(X) = X$ and $\psi(Y) = Y$ and hence $\mathbb{H}(X) + \mathbb{H}(Y) = 0$, which contradicts Theorem 31. □

Lemma 41. *If $G = \mathbb{F}_2^d$ and $\alpha \in (0,1)$ and X, Y are G-valued random variables then there is a subgroup $H \leq \mathbb{F}_2^d$ such that*

$$\log|H| \leq \frac{1+\alpha}{2(1-\alpha)}(\mathbb{H}(X) + \mathbb{H}(Y))$$

and if $\psi : G \to G/H$ is the natural projection then

$$\mathbb{H}(\psi(X)) + \mathbb{H}(\psi(Y)) \leq \frac{20}{\alpha} d[\psi(X); \psi(Y)].$$

Proof. Let $H \leq \mathbb{F}_2^d$ be a maximal subgroup such that

$$\mathbb{H}(\psi(X)) + \mathbb{H}(\psi(Y)) > \frac{20}{\alpha} d[\psi(X); \psi(Y)]$$

and such that there exists $c \geq 0$ with

$$\log|H| \leq \frac{1+\alpha}{2(1-\alpha)}(1-c)(\mathbb{H}(X) + \mathbb{H}(Y))$$

and

$$\mathbb{H}(\psi(X)) + \mathbb{H}(\psi(Y)) \leq c(\mathbb{H}(X) + \mathbb{H}(Y)).$$

Note that this exists since $H = \{0\}$ is an example of such a subgroup or we are done with this choice of H.

We know that G/H is a 2-elementary group and so by Lemma 40 there exists some non-trivial subgroup $H' \leq G/H$ such that

$$\log|H'| < \frac{1+\alpha}{2}(\mathbb{H}(\psi(X)) + \mathbb{H}(\psi(Y))$$

and

$$\mathbb{H}(\psi' \circ \psi(X)) + \mathbb{H}(\psi' \circ \psi(Y)) < \alpha(\mathbb{H}(\psi(X)) + \mathbb{H}(\psi(Y)))$$

where $\psi' : G/H \to (G/H)/H'$. By group isomorphism theorems we know that there exists some H'' with $H \leq H'' \leq G$ such that $H' \cong H''/H$ and $\psi' \circ \psi(X) = \psi''(X)$ where $\psi'' : G \to G/H''$ is the projection homomorphism.

Since H' is non-trivial we know that H is a proper subgroup of H''. On the other hand we know that

$$\log|H''| = \log|H'| + \log|H| < \frac{1+\alpha}{2(1-\alpha)}(1-\alpha c)(\mathbb{H}(X) + \mathbb{H}(Y))$$

and

$$\mathbb{H}(\psi''(X)) + \mathbb{H}(\psi''(Y)) < \alpha(\mathbb{H}(\psi(X)) + \mathbb{H}(\psi(Y))) \leq \alpha c(\mathbb{H}(X) + \mathbb{H}(Y)).$$

Therefore (using the maximality of H) it must be the first condition that fails, whence

$$\mathbb{H}(\psi''(X)) + \mathbb{H}(\psi''(Y)) \leq \frac{20}{\alpha} d[\psi''(X); \psi''(Y)].$$

□

We could use the previous lemma for any value of $\alpha \in (0,1)$, which would give a whole range of estimates in Theorem 44. For definiteness, we specialize only to $\alpha = 3/5$, which gives a constant 2 in the first bound below.

Lemma 42. *If $G = \mathbb{F}_2^d$ and $\alpha \in (0,1)$ and X, Y are G-valued random variables then there is a subgroup $H \leq \mathbb{F}_2^d$ such that*

$$\log|H| \leq 2(\mathbb{H}(X) + \mathbb{H}(Y))$$

and if $\psi : G \to G/H$ is the natural projection then

$$\mathbb{H}(\psi(X)) + \mathbb{H}(\psi(Y)) \leq 34d[\psi(X); \psi(Y)].$$

Proof. Specialize Theorem 41 to $\alpha = 3/5$. In the second inequality, it gives a bound $100/3 < 34$. □

Lemma 43. *Let $\phi : G \to H$ be a homomorphism and $A, B \subseteq G$ be finite subsets. If $x, y \in H$ then let $A_x = A \cap \phi^{-1}(x)$ and $B_y = B \cap \phi^{-1}(y)$. There exist $x, y \in H$ such that A_x, B_y are both non-empty and*

$$d[\phi(U_A); \phi(U_B)] \log \frac{|A||B|}{|A_x||B_y|} \leq (\mathbb{H}(\phi(U_A)) + \mathbb{H}(\phi(U_B)))(d(U_A, U_B) - d(U_{A_x}, U_{B_y})).$$

Proof. The random variables $(U_A \mid \phi(U_A) = x)$ and $(U_B \mid \phi(U_B) = y)$ are equal in distribution to U_{A_x} and U_{B_y} respectively (both are uniformly distributed over their respective fibres). It follows from Theorem 1 that

$$\begin{aligned}\sum_{x,y \in H} \frac{|A_x||B_y|}{|A||B|} d[U_{A_x}; U_{B_y}] &= d[U_A \mid \phi(U_A); U_B \mid \phi(U_B)] \\ &\leq d[U_A; U_B] - d[\phi(U_A); \phi(U_B)].\end{aligned}$$

Therefore with $M := \mathbb{H}(\phi(U_A)) + \mathbb{H}(\phi(U_B))$ we have

$$\left(\sum_{x,y \in H} \frac{|A_x||B_y|}{|A||B|} M d[U_{A_x}; U_{B_y}] \right) + M d[\phi(U_A); \phi(U_B)] \leq M d[U_A; U_B].$$

Since

$$M = \sum_{x,y \in H} \frac{|A_x||B_y|}{|A||B|} \log \frac{|A||B|}{|A_x||B_y|}$$

we have

$$\sum_{x,y \in H} \frac{|A_x||B_y|}{|A||B|} \left(M d[U_{A_x}; U_{B_y}] + d[\phi(U_A); \phi(U_B)] \log \frac{|A||B|}{|A_x||B_y|} \right) \leq M d[U_A; U_B].$$

It follows that there exists some $x, y \in H$ such that $|A_x|, |B_y| \neq 0$ and

$$M d[U_{A_x}; U_{B_y}] + d[\phi(U_A); \phi(U_B)] \log \frac{|A||B|}{|A_x||B_y|} \leq M d[U_A; U_B].$$

□

Theorem 44. *Let $A \subseteq \mathbb{Z}^d$ and $|A - A| \leq K|A|$. There exists $A' \subseteq A$ such that $|A'| \geq K^{-17}|A|$ and $\dim A' \leq \frac{40}{\log 2} \log K$.*

Theorem 45. *If $A, B \subseteq \mathbb{Z}^d$ are finite non-empty sets then there exist non-empty $A' \subseteq A$ and $B' \subseteq B$ such that*

$$\log \frac{|A||B|}{|A'||B'|} \leq 34d[U_A; U_B]$$

such that $\max(\dim A', \dim B') \leq \frac{40}{\log 2} d[U_A; U_B]$.

Proof. Without loss of generality we can assume that A and B are not both inside (possibly distinct) cosets of the same subgroup of $\mathbb{Z}^d$, or we just replace $\mathbb{Z}^d$ with that subgroup. We prove the result by induction on $|A| + |B|$.

Let $\phi : \mathbb{Z}^d \to \mathbb{F}_2^d$ be the natural mod-2 homomorphism. By Theorem 39

$$\max(\mathbb{H}(\phi(U_A)), \mathbb{H}(\phi(U_B))) \leq 10d[U_A; U_B].$$

We now apply Theorem 42, obtaining some subgroup $H \leq \mathbb{F}_2^d$ such that

$$\log|H| \leq 40d[U_A; U_B]$$

and

$$\mathbb{H}(\tilde{\phi}(U_A)) + \mathbb{H}(\tilde{\phi}(U_B)) \leq 34d[\tilde{\phi}(U_A); \tilde{\phi}(U_B)]$$

where $\tilde{\phi} : \mathbb{Z}^d \to \mathbb{F}_2^d/H$ is ϕ composed with the projection onto $\mathbb{F}_2^d/H$.

By Theorem 43 there exist $x, y \in \mathbb{F}_2^d/H$ such that, with $A_x = A \cap \tilde{\phi}^{-1}(x)$ and similarly for B_y,

$$\log \frac{|A||B|}{|A_x||B_y|} \leq 34(d[U_A; U_B] - d[U_{A_x}; U_{B_y}]).$$

Suppose first that $|A_x| + |B_y| = |A| + |B|$. This means that $\tilde{\phi}(A) = \{x\}$ and $\tilde{\phi}(B) = \{y\}$, and hence both A and B are in cosets of $\ker \tilde{\phi}$. Since by assumption A, B are not in cosets of a proper subgroup of $\mathbb{Z}^d$ this means $\ker \tilde{\phi} = \mathbb{Z}^d$, and so (examining the definition of $\tilde{\phi}$) we must have $H = \mathbb{F}_2^d$. Then our bound on $\log|H|$ forces $d \leq \frac{40}{\log 2} d[U_A; U_B]$ and we are done with $A' = A$ and $B' = B$.

Otherwise,

$$|A_x| + |B_y| < |A| + |B|.$$

By induction we can find some $A' \subseteq A_x$ and $B' \subseteq B_y$ such that $\dim A', \dim B' \leq \frac{40}{\log 2} d[U_{A_x}; U_{B_y}] \leq \frac{40}{\log 2} d[U_A; U_B]$ and

$$\log \frac{|A_x||B_y|}{|A'||B'|} \leq 34d[U_{A_x}; U_{B_y}].$$

Adding these inequalities implies

$$\log \frac{|A||B|}{|A'||B'|} \leq 34d[U_A; U_B]$$

as required. □